

UNIT FOUR

MATHEMATICS OF FINANCE

4.1. Interest and its computation

Interest is a fee which is paid for having the use of money. We pay interest on mortgages for having the use of bank's money. Similarly, the bank pays us interest on money invested in saving account.

Interest is usually paid in proportion to the principal and the period of time over which the money is used.

Principal (P) is the amount of money that is lent or invested is called principal. It is the original sum of money.

Interest rater(r) is the rate which specifies interest accumulates. The interest rate is typically stated s a percentage of principal per period of time, for example, 12% per year or 1% per month. Unless otherwise stated, a quoted rate is a rate per year.

Loan (deposit) period (t) is the duration over which the borrowed or deposited money will be used.

Maturity value (amount) (S) is the sum of principal and interest due, which depends on the time of the loan or deposit.

Two categories of interest:

Simple interest

The interest that is paid solely on the amount of the principal is called Simple interest. Simple interest is usually associated with loans or investments which are short-term in nature. The interest is usually considered a single period.

The computation of simple interest is based on the following formula:

Simple interest = principal x interest rate per time period x number of time periods

or

$$I = Prt$$

Note: The time periods for r and t be consistent with each other. That is, if r is expressed as a percentage per year, t should be expressed in number of years. Similarly, if r is expressed as a percentage per month, t must be stated in number of months.

$$\begin{aligned} \text{Maturity value}(S) &= \text{Principal} + \text{interest} \\ &= P + Prt \end{aligned}$$

$$S = P(1 + rt)$$

Example 4.1

Oromia Credit and Saving Union has issued a 3-year loan of Birr 50,000. Simple interest is charged at a rate of 10% per year. The principal plus the interest is to be repaid at the end of the third year. Compute the interest for the 3-year period. What amount will be paid at the end of the third year?

Example 4.2

If Mr.Habtie wanted to buy a taxi which costs Birr 100,000 and if he has only Birr 60,000, what is the amount of interest and the total amount he needs to pay after six months if he considers to borrow the remaining balance at a rate of 12%?

a) Finding time and rate: From the formula of maturity amount, it is possible to derive formula for other variables.

To calculate time

$$S = P(1 + rt)$$

$$\Rightarrow S = P + Prt$$

$$\Rightarrow Prt = S - P$$

$$t = \frac{S - P}{Pr}$$

$$\begin{aligned} \text{OR} \quad (1 + rt) &= S/P \\ rt &= S/P - 1 \end{aligned}$$

$$t = \frac{S/P - 1}{r}$$

To calculate rate:

$$S = P(1 + rt)$$

$$\Rightarrow S = P + Prt$$

$$\Rightarrow Prt = S - P$$

$$r = \frac{S - P}{Pt}$$

$$\begin{aligned} \text{Or} \quad (1 + rt) &= S/P \\ rt &= S/P - 1 \end{aligned}$$

$$r = \frac{S/P - 1}{t}$$

Example 4.3

From a business forecast, you came to know that the price of a Toshiba lap top will be Birr 6000 after 4 years from now. You want to buy 4 lap tops by then. If you have Birr 15,000 currently, at what rate should you agree while depositing to meet your objective?

Example 4.4

From a business forecast, you came to know that the price of a Toshiba lap top will be Birr 6000 in the following 6 years from now. But you want to buy as soon as you get sufficient money to buy 4 lap tops at the same time. If you have Birr 15,000 currently, how long it will take you if you deposit the money you have in Awash Bank at a rate of 15% simple interest to meet your objective?

b) Simple interest when “t” is expressed in days

When time for the maturity period for the loan or deposit is given in days, there are two ways of computing interest: Exact method and ordinary method.

If exact method is used, then the time is:

$$t_e = \frac{\text{number of days}}{365}$$

If ordinary method is used,

$$t_o = \frac{\text{number of days}}{360}$$

Ordinary method is also called Banker’s method. Banks for convenience, often count a year as 12 months in a year, 30 days in a month and 360 days in a year.

Example 4.5.

If you have a short run note borrowed on June 24, 1999 and will be paid on Sep 27, 1999, compute the number of days for which you will pay interest using:

- i) exact method
- ii) Ordinary method

Example 4.6.

If a loan of Birr720 at 10% simple interest was made on January 22, 1976 which is to be paid on March 22, 1976, what is the interest to be paid using:

- i) exact method
- ii) ordinary method

c) Finding Principal or present value

Present value is the value at the time of investment such as principal. The process of finding the present value of a given amount which is due in a future date is called discounting at simple interest.

$$\begin{array}{c} \text{P interest} \rightarrow S \\ \leftarrow \text{Discount} \end{array}$$

Here both the interest and discount are equal in magnitude, but not in all cases.

The formula to calculate present value still can be derived from the maturity value formula, i.e,

$$S = P(1 + rt)$$

$$P = \frac{S}{(1 + rt)}$$

Example 4.7

How much money Ato Assegid should invest today at 6% simple interest if he is to receive Birr 1416 as an amount in 3 years?

- i) What is simple discount?
- ii) What is the present value after one year?
- iii) What is the present value after two year?

Compound Interest

In compound interest, the interest of succeeding period is computed not only based on the original loan or investment, but also on the interest of the previous period. i.e The interest is calculated from both the principal and interest.

Two observations regarding compound growth are:

- the value of an investment can be increased significantly if interest is reinvested as it is earned

- the effect of compound interest are magnified as the interest rate increases.

The frequency of compounding interest has an effect on the value of an investment. Frequency refers to how often interest is computed and earned. The frequency of compounding usually range “annual” compounding to continuous compounding. The value of investment increases with the frequency of compounding.

- The larger the frequency of compounding, the greater the interest to be earned from investment.
- The higher the interest rate, the greater the interest earned each compounding period and the greater the rate of growth in the investment.

Compounding period (interest period) (n) refers to the number of periods on which interests are computed throughout the term of the loan. It is also called conversion period.

Number of conversion period (m) refers to the number of times the interest is calculated on the deposited investment per year.

Nominal rate(r)- refers to annual(rate per annum) or quoted rate for that year unless there are qualifiers like 1.5% per month.

Interest per conversion period (i) - refers to the interest for particular conversion period.

From the above definitions, we can derive a formula for:

$$n=mt \text{ and}$$

$$i=r/m$$

Example 4.8

Assume that you have deposited Birr 8,000 in a credit union which pays interest of 8% per year compounded quarterly. Determine the amount of money that you will have in a deposit at the end of one year if all interest is left in the saving account?

What would be the simple interest for the year and compare the result with that of compound interest?

Compound Amount

If an amount of money P earns an interest compounded at a rate of i percent per period, it will grow after n periods to the compound amount S, which is the future value of the money, where

$$S=P(1+i)^n \quad \text{..Compound interest formula}$$

OR

$$S=P(1+r/m)^{mt} \quad \text{where,}$$

S= compound amount

P= principal

i=interest per compounding period = r/m

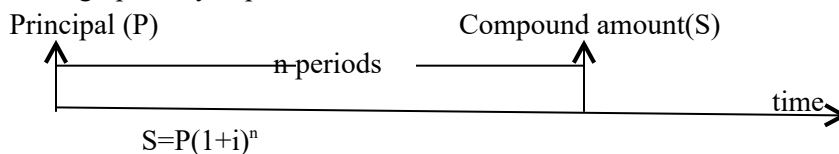
n= #of conversion periods =mt

m= # of conversion times

Compound amount formula

- Compound amount after one period= $P(1+i)$
- Compound amount after two periods= $P(1+i)^2$
- Compound amount after three periods= $P(1+i)^3$...

This can be graphically depicted as:



Example 4.9

Assume you and your colleague want to make an investment of Birr 250,000. The interest rate is 12% per year, and interest is compounded quarterly. If all interest is reinvested at the same interest rate, what will be the value of investment after 8 years?

Present Value

Present value refers to the amount of money that is to be initially invested or deposited to collect certain sum in the future.

$$S = P (1+i)^n$$

The compound amount formula is an equation involving four variables: S, P, i, and n. Given the values of any three of these four, the equation can be solved for the remaining variables.

$$P = \frac{S}{(1+i)^n} = S \left(\frac{1}{(1+i)^n} \right) = S (1+i)^{-n}$$

↑ Present value factor

Example 4.10

Suppose that a person can invest money in a saving account at the rate of 10% per year compounded quarterly. Assume that the person wishes to deposit a lump sum at the beginning of the year and have that sum grow to Birr 20,000 over the next 10 years. How much money should be deposited? **Computing Time and interest rate**
Compound interest calculations involve exponents and, as a consequence, logarithms play a role in some of these calculations.

Some rules of logarithms which are relevant to this discussion are listed as follows:

1. $\log b^m = \log^m + \log^n \Rightarrow$ It is possible not to write the base if it is to the base 10.
2. $\log b^{m/n} = \log b^m - \log b^n$
p
3. $\log b^m = p \log b$
4. $\log a^y = \frac{\log b^y}{\log b^a}$
5. $\log b^b = 1$
x
6. $\log b^b = x$
7. $\text{Antilog}(\log^x) = x$
8. $\text{Antilog}(\log)^x = x$

Example 4.12

How long it will take to accumulate Birr 650 if Birr 500 is invested at 10% compounded quarterly?

Example 4.13

At what rate should you deposit Birr 500 compounded quarterly to accumulate Birr 650 in 2.66 years

Effective interest Rates

Because of lack of comparability, it is hard to judge whether interest quoted at 8% compounded semiannually results in more or less than would be in the case if the rate was 7.9% compounded monthly. To make the comparison possible, we change both to their equivalent annual rates; these equivalent rates are called effective rates.

Interest rates are typically stated as annual percentages. The stated annual rate is referred to as nominal rate. When interest is compounded semiannually, quarterly, and monthly, the interest earned during a year is greater than if compounded annually. When compounding is done more frequently than annually, an effective annual interest rate can be determined. This is the interest rate compounded annually which is equivalent to a nominal rate compounded more frequently than annually. The two rates would be considered equivalent if both result in the same compound amount.

Example 4.14

Birr 1 at 8% compounded quarterly for a year would amount to:

$$S = 1(1+0.08/4)^4$$

$$= (1.02)^4$$

= 1.08243, which is the same as the amount of Birr 1 at a rate of 0.08243, or 8.243% for one year.

Similarly, Birr 1 at 7.9% compounded monthly for a year would amount to:

$$S=1\left(1+\frac{(0.079)}{12}\right)^{12}=1.08192, \text{ which is equivalent to the amount of Birr 1 at a rate of } 8.192\% \text{ for one year.}$$

From the above example, 1.08192 is composed of 1 and effective rate=0.08192.

From this, a nominal (annual) rate i compounded m times a year, birr 1 grows to:

$$S = (1+i_e), \text{ where } i_e \text{ is effective rate.}$$

$$(1+i)^m = (1+i_e), \text{ here } n=m, \Rightarrow n=mt \Rightarrow n = m \times 1 \text{ year} \Rightarrow n = m$$

Therefore,

$$i_e = (1+i)^m - 1 \quad \text{where } i=r/m$$

Example 4.15

Suppose you read in a business news paper that stocks sold by Oromia international Bank pay 15% compounded monthly, and Stock sold by Bunna international Bank pay 15.2% compounded semiannually. Which of the stocks would you buy assuming citrus puribus?

4.2. Annuities

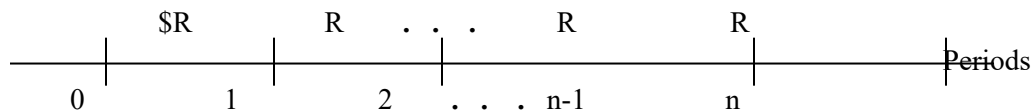
Annuity is a series of periodic payments usually made in equal amounts. The deposits or payments are made periodically like semiannually, quarterly, monthly, etc. Examples of annuities include:

- Regular deposit to a saving account by an employee
- Regular payment to mortgage
- Periodic payment to cover loan for car purchase
- Monthly payment to government employee from a retirement fund
- Regularly payment of insurance premium to insurer
- Regular tax payment by self-employed person.
- Others

Payment interval-refers to the period of time between successive payment dates. The time interval between the beginning of the first payment and the end of the last payment is called term of the annuity.

Ordinary Annuity

Ordinary Annuity refers to a series of equal periodic payments, in which each payment is made at the end of compounding period.



n =number of periods compounding

i =interest rate per compounding period

R =periodic payment

S =Future value of the annuity

In the above figure, $\$R$ accumulates interest for $n-1$ periods, the second $\$R$ for $n-2$ periods and etc. The next to last payment $\$R$ accumulates one period of interest and the last payment $\$R$ accumulates for no interest. So, using future value formula for compounding interest, future value of ordinary annuity will be:

$$S = R(1+i)^{n-1} + R(1+i)^{n-2} + \dots + R(1+i)^2 + R(1+i)^1 + R \dots (1)$$

[$(S = R(1+i)^{n-1} + R(1+i)^{n-2} + \dots + R(1+i)^1 + R) (1+i)$] ...multiply both sides by $(1+i)$ which gives:

$$S(1+i) = R(1+i)^{n-1}(1+i) + R(1+i)^{n-2}(1+i) + \dots + R(1+i)^1(1+i) + R(1+i)$$

$$S(1+i) = R(1+i)^n + R(1+i)^{n-1} + \dots + R(1+i)^2 + R(1+i) \dots (2)$$

Subtract (1) from (2)

$$S(1+i) = R(1+i)^n + R(1+i)^{n-1} + R(1+i)^{n-2} + \dots + R(1+i)^2 + R(1+i)^1 + R(1+i)^0$$

$$-(S = R(1+i)^{n-1} + R(1+i)^{n-2} + \dots + R(1+i)^2 + R(1+i)^1 + R(1+i)^0 + R)$$

$$\Rightarrow S(1+i) - S = R(1+i)^n - R$$

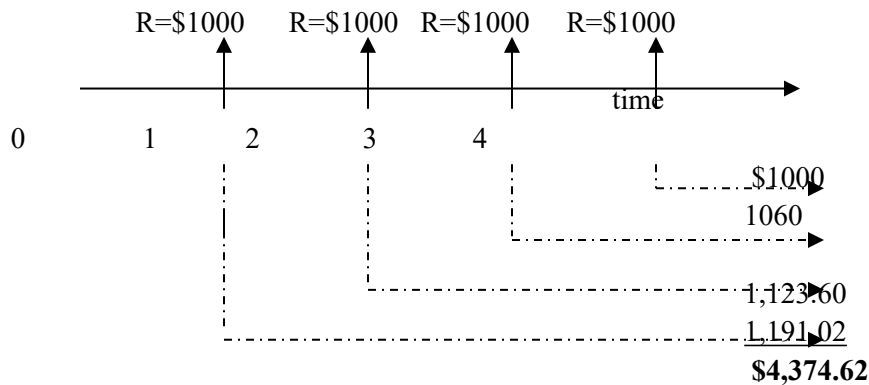
$$\Rightarrow S + Si - S = R \left[\frac{(1+i)^n - 1}{i} \right]$$

$$\Rightarrow Si = R \left[\frac{(1+i)^n - 1}{i} \right]$$

$$S = \frac{R \left[\frac{(1+i)^n - 1}{i} \right]}{i} \quad \text{Future value of ordinary annuity}$$

Example 4.18

A person plans to deposit \$1000 in a saving account at the end of this year and equal sum at the end of each following year. If interest is expected to be earned at the rate of 6% per year compounded annually, to what sum will the investment grow at the time of the fourth deposit?



Note that the first deposit earns interest for three years, although the fourth deposit earns no interest. Of the \$4,374.62, the interest which has been earned on the first three deposits is \$374.62.

The procedure used to determine in the above example is manageable, but impractical when the number becomes large. The simpler approach is to use the formula derived above for S,

$$S = R \left[\frac{(1+i)^n - 1}{i} \right] \dots \text{Future value of ordinary annuity}$$

Special symbol $S_{n,i}$, which verbalized as “s sub n angle i,” is frequently used to abbreviate the series compound amount factor. Thus, the above equation can be rewritten more simply as:

$$S = R \left[\frac{(1+i)^n - 1}{i} \right]$$

$$= 1000 \left[\frac{(1+0.06)^4 - 1}{0.06} \right] \quad \text{Using scientific calculator,}$$

$$= \$4,374.62$$

Example 4.19

Birr 20 is deposited at the end of each quarter for 5 years in Commercial Bank of Ethiopia, which pays 12% interest compounded quarterly. If CBE decide to change its nominal rate to 16% compounded quarterly after 3 years, what will be the amount of the annuity at the end of the 5 years term and what is the total interest that is earned?

Note: it is not only the interest rate which changes, but also the amount of periodic deposits does.

Example 4.19

An employee deposits Birr 150 at the end of each month in a credit union saving plan that pays an interest rate of 12% compounded monthly. If the monthly periodic payments have increased to Birr 200 as a result of salary increment after the 8th payment, what is the balance after one year? What is the interest that is earned?

Sinking Funds

A sinking fund is a fund in to which periodic payments are made in order to accumulate a specified amount at some point in the future. It involves determining the size of an annuity. It answers, what amount should be deposited each period in order to reach the goal? The formula to calculate the size of annuity, R is determined from the future value of the annuity.

$$R = S \left(\frac{1}{\frac{(1+i)^n - 1}{i}} \right) \quad R = S \left(\frac{i}{(1+i)^n - 1} \right)$$

↑ This **expression** is the reciprocal of the series of compound amount factor.

This factor is often called the sinking fund factor. This is because the series of deposits used to accumulate some future money is often called a sinking fund. If you use financial table, you will go to where the sinking fund factor exists and then find the value at the intersection of n and i. Then, multiply the result by S.

Example 4.20

Kidist wants to settle a debt of Birr 12,000 to be paid 10 years from now. If deposit earn interest at a rate of 5% compounded annually, how much money should she deposit at the end of each year for coming 10 years and what is the amount of interest earned?

Example 4.21

A loan Birr 15,000 with interest at 10% semiannually must be paid off in one payment after three years. In order to settle this amount, quarterly payments are to be placed in a sinking fund which pays 16% compounded quarterly. How much should each quarterly payment be?

Present Value of ordinary annuity

Present value of ordinary annuity is a single principal which at a given compound interest rate will accumulate to the amount of annuity by the end of the term of the loan. In other words, it is an amount of money today which is equivalent to a series of equal payments in the future. The deposits will be discounted for n periods.

Dear learners, from our earlier discussions, present value of any amount due n periods from now is found by multiplying the amount by the compound discount factor, $(1+i)^{-n}$.

If we multiply the future amount of an annuity of Birr 1 per period by the compound discount factor, we have the present value of an annuity of Birr 1 per period.

Hence, $P = S_n (1+i)^{-n}$

$$P = R \left(\frac{(1+i)^n - 1}{i} (1+i)^{-n} \right)$$

$$P = R \left(\frac{(1+i)^n (1+i)^{-n} - 1(1+i)^{-n}}{i} \right)$$

$$P = R \left(\frac{(1+i)^0 - 1(1+i)^{-n}}{i} \right)$$

$$P = R \left(1 - \frac{(1+i)^{-n}}{i} \right) \text{ Present value of Ordinary Annuity}$$

Present value annuity calculations arise when we wish to determine what lump sum must be deposited in an account now if this sum and the interest it earns are to provide equal payments for a stated number of periods, with the last payment making the account balance zero.

Example 4.22

Ato Abebe recently won a national lottery. The terms of the lottery are that the winner will receive annual payments of Birr 20,000 at the end of this year and each of the following three years. If the winner could invest money today at the rate of 8% per year compounded annually, what is the present value of the four payments?

Example 4.23

Parents of a teenage girl want to deposit a sum of money which will earn interest at the rate of 9% per year compounded semiannually. The deposit will be used to generate a series of eight semiannual payments of Birr 2500 beginning 6 months after the deposit. These payments will be used to help finance their daughter's college education. What amount must be deposited to achieve their goal? How much interest will be earned on this deposit?

Amortization

Amortization involves the periodic payment of a current obligation discharged by making a series of payments in the future. After the last payment, the obligation ceases to exist – it is dead-and it is said to have been amortized by the payment. The process of repaying a loan by installment payments is referred to as amortizing a loan. Prominent examples of amortization are loans taken to buy a home or a car and amortized over a period of 20 to 30 years. From the present value of annuity formula, we can derive for R, as:

$$R = \left(\frac{P}{\frac{1-(1+i)^{-n}}{i}} \right) \Rightarrow R = P \left(\frac{i}{1-(1+i)^{-n}} \right), \text{ Here the problem is to determine the size of an annuity.}$$

Example 4.24

Samuel borrowed Birr 5000 to buy a car. He will amortize the loan by monthly payments over three years.

- Find the monthly payment if the interest is 12% compounded monthly?
- Find the total amount that Samuel will pay?

Example 4.25

A company borrowed Birr 50,000 at 10% compounded quarterly. The debt is to be amortized by equal payments each quarter over 15 years. Calculate the quarterly payment and the interest to be paid?

Mortgages

In typical home purchase transactions, the home owner pays part of the cost in cash and borrows the remainder needed, usually from a bank or a savings, or loan institution. The buyer amortizes the indebtedness by periodic payments over a period of time. As a security for the loan, the lending organization ordinarily obtains conditional title to the property, **mortgage**. Given a mortgage loan, many home owners do not realize how the amount of their mortgage payment is calculated. It is calculated in the same way as were for the amortization in the last section.

Hence,

$$R = P \left(\frac{i}{1-(1+i)^{-n}} \right)$$

However, interest is typically compounded monthly, time period is long, and interest rate per compounding period can equal unusual fractions or decimal answers. Because large sums of money are involved, amortization calculations require carrying large decimals.

Example 4.27

A person pays Birr 100,000 for a new house. A down payment of Birr 30,000 leaves a mortgage of Birr 70,000, with interest computed at 10.5% per year compounded monthly. Determine the monthly mortgage payment if the loan is to be repaid over (a) 20 years (b) 25 years (c) 30 years (d) compute the interest under the three different loan periods.

Amortization Schedule

When a debt is amortized, part of each payment is interest on the balance outstanding, and the remainder is used to reduce the balance outstanding. The largest interest charge occurs at the first payment because then interest is due on the entire principal and, of course, the smallest reduction of principal occurs at the time of the first payment. To see how this works, recall the Birr 70,000, 10.5%, 20 year mortgage of the above example. The monthly payment is Birr 698.87. At the time the first payment is due, one month has passed and the interest on Birr 70,000 is

$$I = 70,000 \times 10.5\% \times 1/12 = \text{Birr } 612.5$$

The amount applied to reduce the balance outstanding is:

$$\text{Birr } 698.87 - \text{Birr } 612.5 = 86.37,$$

So the new balance is Birr 70,000 - 86.37 = Birr 69,913.63

Continuing to the second payment, the beginning balance is now Birr 69,913.63, and we proceed as follows:

Beginning balance owed		Birr 69,913.63
Payment	Birr 698.87	
Interest charge $(69,913.63)(0.105)(1/12)$	<u>611.74</u>	
Reduction in balance owed	87.13	<u>87.13</u>
Ending balance owed		Birr 69,826.50

From the above calculations, we can form amortization schedule table as follows:

	(1) P	(2)	3(Prt)	4(2-3)	5(1-4)
Month	Beginning Balance owed	Monthly Payment	Interest charge	Reduction in balance owed	Ending balance owed
1	70,000	698.87	612.5	86.37	69,913.63
2	69,913.63	698.87	611.74	87.13	69,826.50
3	69,826.50	698.87	610.98	87.89	69,738.61
4	69,738.61	698.87	610.21	88.66	69,649.95
5	69,649.95	698.87	609.44	89.43	69,560.52
6	69,560.52	698.87	608.65	90.22	69,470.3
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
240	...	...		...	Zero

Amortization Schedule

☞ Note: Each month, the interest charge decreases and the reduction of the amount owed increases. Note also that this process is so tedious especially when n becomes larger. In such cases, computers can be employed to perform the task easily.